

A PROJECT REPORT ON

Customer Churn Analysis and Predictive Reporting Using Machine Learning and Power BI

B.Sc. Data Science (Sem IV)

SUBMITTED BY

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UNDER THE GUIDANCE OF

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SCHOOL OF DATA SCIENCE

SYMBIOSIS SKILLS AND PROFESSIONAL UNIVERSITY, PUNE

ACADEMIC YEAR 2025-26

CERTIFICATE

This is to certify that

*Dipannita Pramanik, Sanjivani Sathe, Kanak Ghanekar, Anant Singhal
and Gouri Juvekar*

*have successfully completed the
Project-Based Learning Report in Data Science*

titled

Customer Churn Analysis and Predictive Reporting

Using Machine Learning and

Power BI

during the academic year 2025-26.

*The project work was carried out under the guidance of
Dr. Abhijeet Thakur*

and is approved as per the academic requirements.

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DECLARATION

I hereby declare that the project report titled “**Customer Churn Analysis and Predictive Reporting Using Machine Learning and Power BI**” submitted in partial fulfillment of the requirements for the award of the degree in **B.Sc. Data Science** is a record of original work carried out by me/us during the academic year **2025-26**.

I/We further declare that this project has not been submitted to any other university or institution for the award of any degree, diploma, or certificate.

The work presented in this report is genuine and has been carried out under the guidance of **Dr. Abhijeet Thakur**. All sources of information used in this project have been properly acknowledged and cited.

I/We understand that any act of plagiarism or academic dishonesty, if found, will result in appropriate disciplinary action as per institutional norms.

Place: Pune

Date: 15-04-2026

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CHAPTER 1: INTRODUCTION

1.1 Overview

In the modern banking sector, retaining customers is as critical as acquiring new ones. Customer Churn refers to the phenomenon where customers stop doing business with an organization — in banking, this means closing accounts or ceasing all transactions. This project focuses on Bank Customer Churn Analysis, where demographic and financial data of 10,000 customers is analyzed to identify patterns that lead to account closures.

By integrating MySQL for structured data management, Machine Learning for predictive modeling, Power BI for visual analytics, and Google Gemini AI for strategic recommendations, this project provides a complete end-to-end solution for proactive customer retention in the banking sector.

1.2 Problem Statement

The primary challenge addressed in this project is that many banks lack automated tools to predict which customers are likely to leave. The core problems identified are:

- **Data Silos:** Customer data is stored in relational databases but is not effectively utilized for forward-looking predictions
- **Reactive Approach:** Traditional systems flag customers only after they have already churned — there is no early warning mechanism
- **Lack of Actionable Insights:** Raw data must be converted into visual analytics that non-technical bank managers can actually use to make decisions

1.3 Objectives

The primary objectives of this Project Based Learning (PBL) are:

- To develop a robust data pipeline using MySQL for structured storage and retrieval of customer records
- To build and compare four Machine Learning models — Logistic Regression, Decision Tree, Random Forest and XGBoost — to predict customer churn probability
- To handle class imbalance using SMOTE to ensure fair and unbiased prediction of the minority churned class
- To create an interactive 3-page Power BI dashboard for executive level reporting and visual analytics
- To integrate Google Gemini AI API to generate personalized retention strategies for at-risk customers
- To deploy a real time prediction web application using Streamlit

1.4 Scope of the Project

The scope of this project extends beyond traditional statistical reporting to provide an intelligent decision support system. It includes:

- **Data Integration:** Reliable connection to MySQL relational database handling 10,000 customer records
- **Predictive Analytics:** High performance ML algorithms computing churn probabilities for every customer using all input features together
- **Class Imbalance Handling:** SMOTE applied to correct the 3.91:1 imbalance ratio in the dataset
- **AI Integration:** Google Gemini API transforming raw probability scores into human-readable personalized retention strategies
- **Executive Visualization:** Interactive Power BI dashboard allowing stakeholders to segment customers by risk, geography and financial behavior
- **Web Deployment:** Streamlit based ChurnAI web application for real time individual customer churn assessment

CHAPTER 2: LITERATURE REVIEW/ BACKGROUND STUDY

2.1 Background of Customer Churn

Customer churn is a critical business challenge faced by organizations across service-based industries, especially banking. Churn occurs when a customer stops using a company's services — in banking this means closing accounts or ceasing all transactions. Understanding why customers leave and predicting who is likely to leave next is essential for sustainable business growth.

Traditionally, banks relied on end-of-month reports that listed customers who had already exited. This was a purely reactive approach — the bank could only respond after the damage was done. There was no system in place to identify at-risk customers before they made the decision to leave.

2.2 Need for Predictive Analytics

With growing competition in the banking sector, organizations realized that retaining an existing customer is far less expensive than acquiring a new one. This created a strong need for a **proactive** approach — one that could analyze customer behavior patterns and predict churn before it happens.

This is where Machine Learning and Data Science play a vital role. By analyzing customer attributes such as age, account balance, number of products, activity status and geography, a predictive model can assign a churn risk score to every customer — giving the bank enough time to intervene with personalized retention strategies.

2.3 Role of Each Technology in Churn Analysis

Database System (RDBMS) Customer data is typically stored across multiple tables in relational databases. MySQL provides a structured and reliable way to store, manage and retrieve large volumes of customer records. SQL queries allow efficient extraction of relevant data for analysis. In this project, all 10,000 customer records are stored in a MySQL database and fetched directly into the Python ML pipeline.

Machine Learning Machine Learning algorithms are the core of any churn prediction system. Supervised learning techniques — specifically binary classification — are used to learn patterns from historical data and predict whether a new customer will churn or stay. Models like Logistic Regression, Decision Tree, Random Forest and XGBoost each bring different strengths to the prediction task, and comparing them ensures the most reliable model is selected.

Power BI Raw predictions from ML models are difficult for non-technical business managers to interpret. Power BI bridges this gap by transforming complex numerical outputs into interactive visual dashboards. KPI cards, bar charts, donut charts and scatter plots allow stakeholders to understand churn trends across geography, age groups, product usage and activity levels — without any coding knowledge.

Advanced Business Domain / CRM Customer Relationship Management (CRM) is the business strategy layer of this project. The insights generated by ML models and visualized in Power BI are translated into actionable retention strategies. For example, identifying that Germany has the highest churn rate or that inactive members are most at risk allows bank managers to design targeted loyalty programs and personalized intervention campaigns.

2.4 Research Gap Identified

Most existing churn analysis projects are limited to just building a model in Python or just creating a dashboard — they do not provide a complete integrated solution. Key gaps identified include:

- No connection between the database, ML model and visualization tool
- Class imbalance in churn datasets is often ignored leading to biased models
- Predictions are presented as numbers without any business context or actionable strategy
- No real time interface for bank employees to assess individual customer risk

2.5 How This Project Addresses the Gap

This project addresses all the above gaps by delivering a complete end-to-end system:

- **MySQL → Python** connection for structured data retrieval
- **SMOTE** applied to handle class imbalance and ensure unbiased predictions
- **Power BI dashboard** to present insights visually to non-technical stakeholders
- **Streamlit web application** for real time individual customer churn prediction
- **Google Gemini AI** to convert numerical predictions into human readable personalized retention strategies

This integrated approach transforms raw banking data into a proactive customer retention tool — directly aligned with the project's problem statement of understanding churn patterns for business growth.

CHAPTER 3: PROBLEM DEFINITION

3.1 Problem Statement

Customer churn in the banking industry is a complex problem because it involves an interplay between financial behavior, service satisfaction and competitive market alternatives. When a customer exits a bank, the institution does not just lose a single account — it loses the potential for cross-selling insurance, loans and investment products, representing a significant blow to Customer Lifetime Value (CLV).

This project defines the problem as a **binary classification task** where the goal is to predict whether a customer will Exit (1) or Stay (0) based on their demographic and financial profile. The challenge is further complicated by class imbalance — only 20.37% of customers in the dataset actually churned, making it difficult for standard models to identify the minority class accurately.

3.2 Dataset Description

The dataset used is the **Bank Customer Churn Modelling dataset** sourced from Kaggle and stored in a MySQL database. It contains **10,000 customer records** with the following attributes:

Table 3.1: Dataset Description

Column	Type	Description
CreditScore	Numerical	Customer's credit score
Geography	Categorical	Country — France, Spain, Germany
Gender	Categorical	Male or Female
Age	Numerical	Customer's age
Tenure	Numerical	Years with the bank
Balance	Numerical	Account balance
NumOfProducts	Numerical	Number of bank products used
HasCrCard	Binary	Has credit card (1 = Yes, 0 = No)
IsActiveMember	Binary	Active member (1 = Yes, 0 = No)
EstimatedSalary	Numerical	Estimated annual salary
Exited	Binary	Target — 1 = Churned, 0 = Stayed

Key Statistics:

- Total Records: 10,000
- Churned Customers: 2,037 (20.37%)
- Non-Churned Customers: 7,963 (79.63%)
- Class Imbalance Ratio: 3.91:1
- Missing Values: None

CHAPTER 4: SYSTEM DESIGN & METHODOLOGY

4.1 System Architecture

The architecture of this project is designed as a multi-tier pipeline ensuring seamless data flow from storage to end-user insights. It consists of four primary layers:

1. **Data Layer** — MySQL relational database serves as the source of truth storing all 10,000 customer records
2. **Processing Layer** — Python scripts handle data cleaning, feature engineering and execution of all ML models
3. **Intelligence Layer** — Google Gemini AI API interprets churn probabilities and generates human readable retention strategies
4. **Presentation Layer** — Dual interface approach using Power BI for executive dashboards and Streamlit for real time web predictions

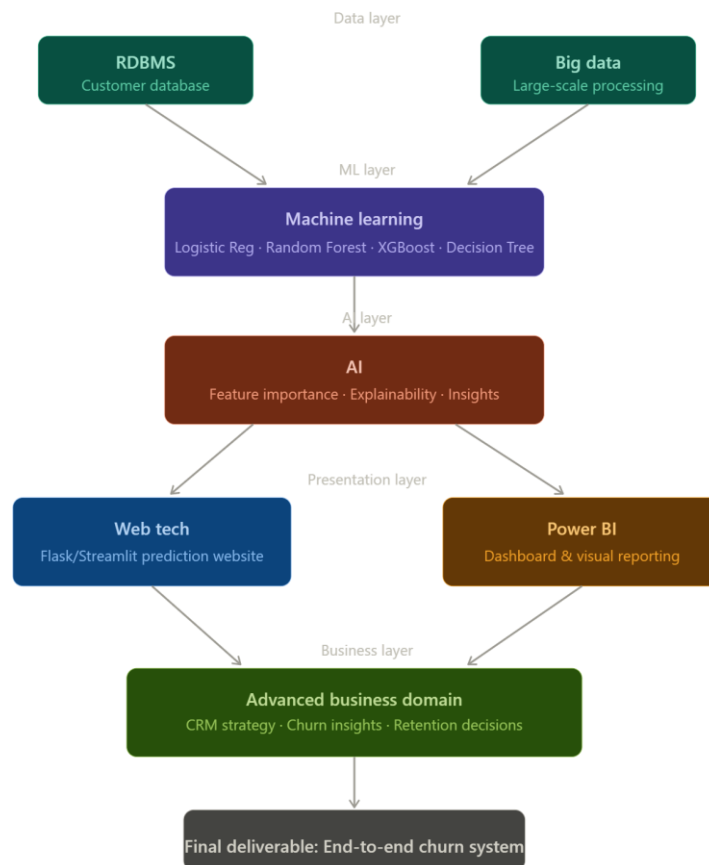


Figure 4.1: System Architecture for Predictive Churn Analysis

4.2 Tools & Technologies Used

Table 4.2: Tools and Technologies

Tool / Technology	Purpose
MySQL	Relational database for storing customer records
Python 3.x	Backend programming language
Pandas & NumPy	Data manipulation and numerical analysis
Scikit-learn	Preprocessing, model building and evaluation
Imbalanced-learn	SMOTE for handling class imbalance
XGBoost	Gradient boosting classification
Joblib	Saving and loading trained models
Streamlit	Web application development
Google Gemini API	AI powered retention strategy generation
Power BI Desktop	Interactive dashboard and visual reporting

4.3 Project Workflow / Block Diagram

The project follows a logical 9-step workflow from data ingestion to final output:

1. Connect to MySQL and fetch customer data
2. Data cleaning — drop irrelevant columns, check for nulls
3. Exploratory Data Analysis — visualize patterns and distributions
4. Encode categorical variables and scale features
5. Handle class imbalance using SMOTE
6. Train four ML models
7. Compare and evaluate models
8. Feature importance analysis
9. Save best model as .pkl file for web deployment

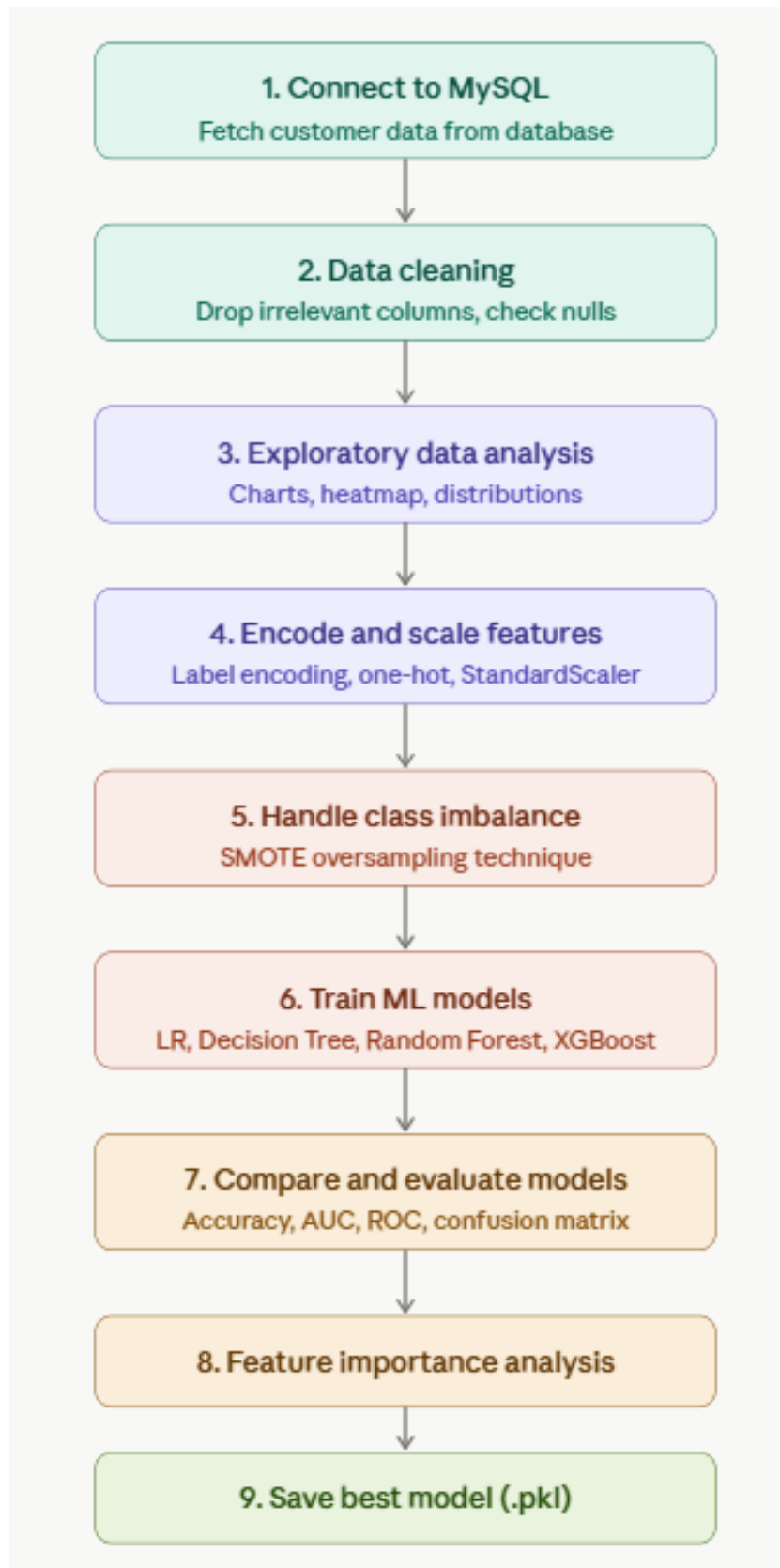


Figure 4.2 Project Workflow

4.4 Algorithms Used

Logistic Regression A statistical method for binary classification that models the probability of churn using a sigmoid function. Selected as the baseline model due to its simplicity and interpretability.

Decision Tree A tree based classifier that splits data based on feature conditions. Easy to visualize but prone to overfitting on complex datasets.

Random Forest An ensemble of 100 decision trees using bagging. Final prediction is made by majority voting across all trees — reduces variance and overfitting significantly.

XGBoost An advanced gradient boosting algorithm that builds trees sequentially where each tree corrects errors of the previous one. Highly efficient and accurate for tabular classification tasks.

CHAPTER 5: DATA COLLECTION & PREPROCESSING

5.1 Data Source

The dataset was sourced from **Kaggle** (Bank Customer Churn Modelling) and imported into a **MySQL database** named `churn_db` under the table `customers`. Python connects to this database using the `mysql-connector-python` library and fetches all 10,000 records directly into a Pandas DataFrame — replacing the traditional approach of reading from a CSV file.

5.2 Exploratory Data Analysis (EDA)

Before preprocessing, an extensive EDA was performed to understand the data distribution and relationships between features and churn:

Churn Distribution — 79.63% customers stayed, 20.37% churned confirming class imbalance

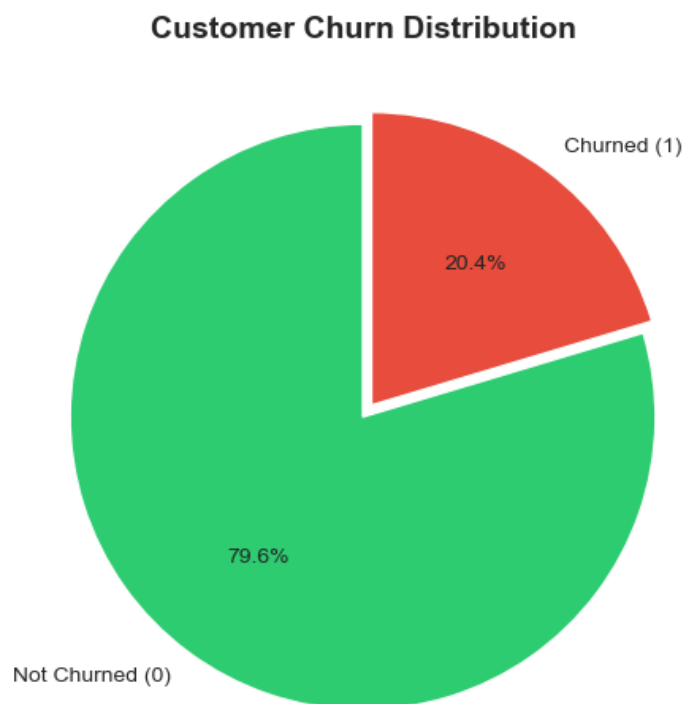


Figure 5.1 Churn Distribution

- **Churn by Geography** — Germany showed notably higher churn compared to France and Spain

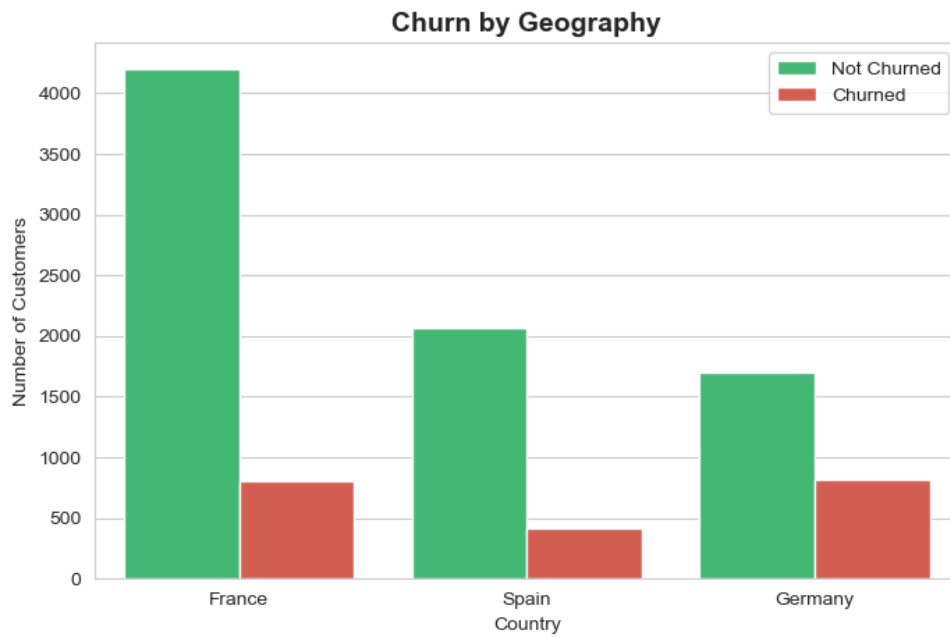


Figure 5.2 Churn by Geography

- **Churn by Gender** — Female customers showed higher churn tendency than male customers

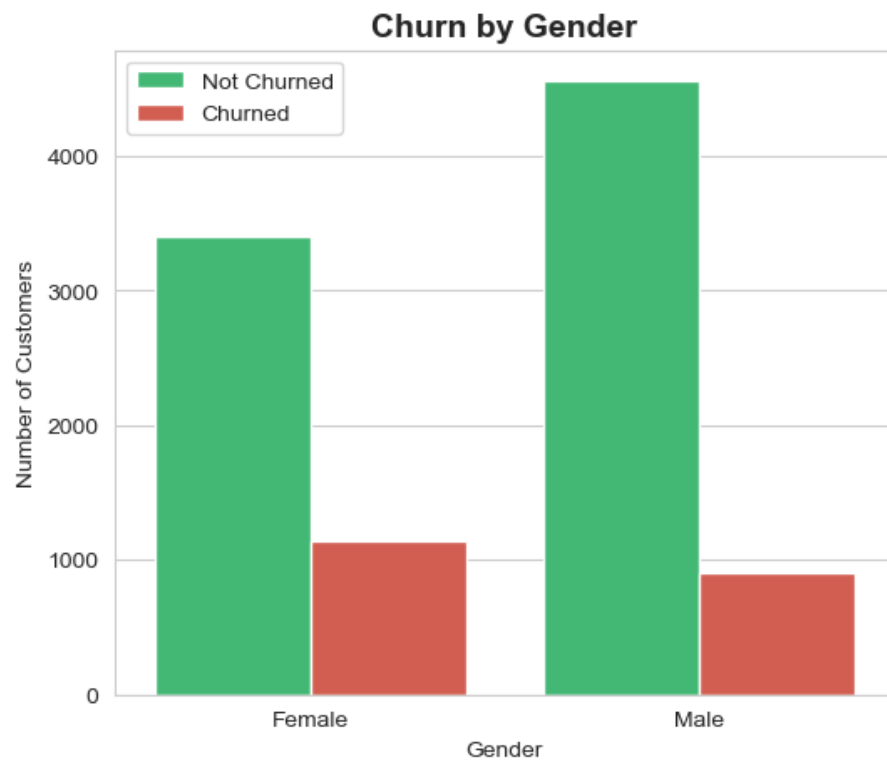


Figure 5.3 Churn by Gender

- **Age vs Churn** — Customers aged 40-60 are significantly more likely to churn

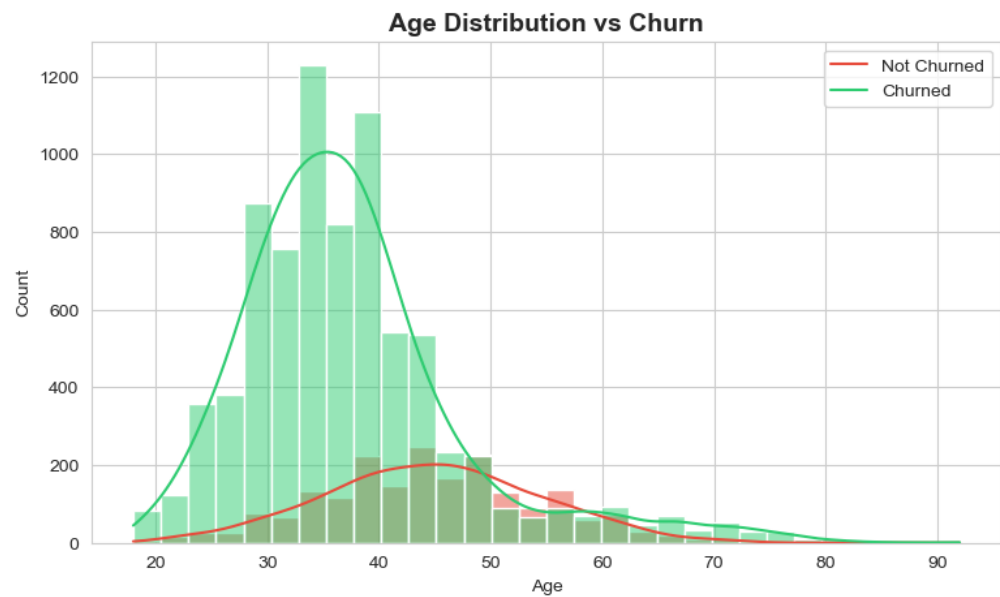


Figure 5.4 Age Vs. Churn

- **Balance vs Churn** — Customers with higher account balances paradoxically showed higher churn risk

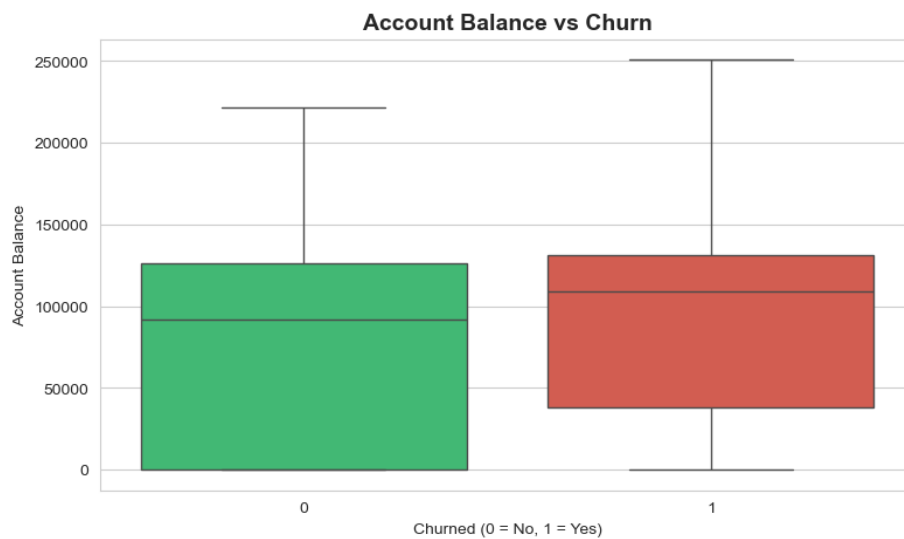


Figure 5.5 Balance Vs. Churn

- **Correlation Heatmap** — Age and IsActiveMember showed the strongest correlation with the Exited target variable

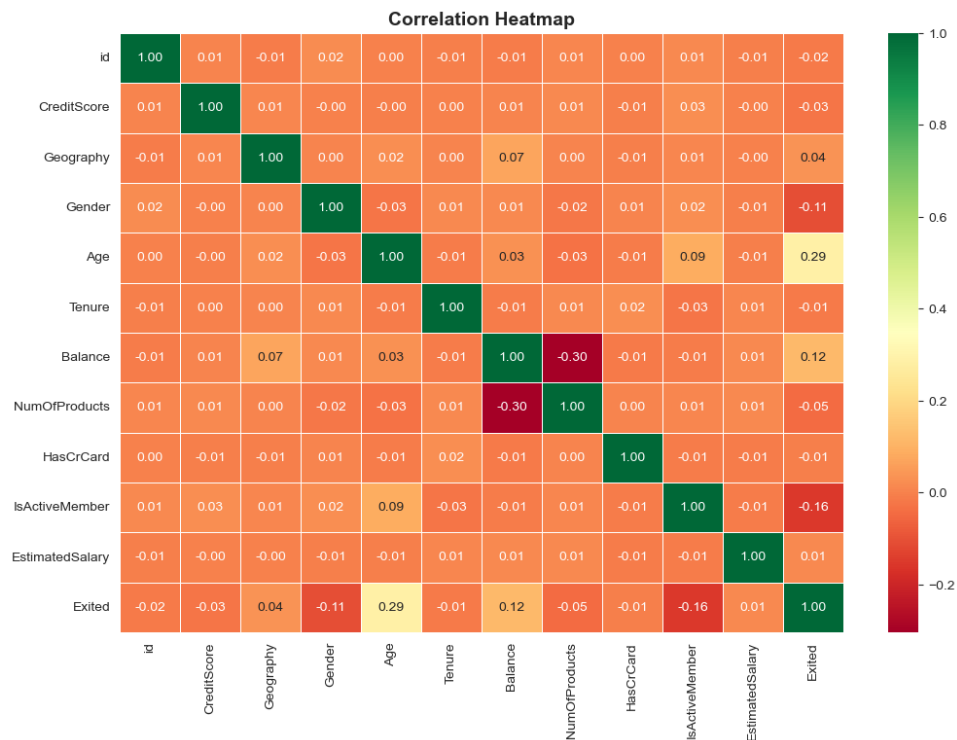


Figure 5.6 Correlation Heatmap

5.3 Data Cleaning

The following cleaning steps were performed:

- **Dropping Irrelevant Columns:** RowNumber, CustomerId and Surname were removed as they are unique identifiers that do not contribute to churn prediction and would cause the model to overfit
- **Null Value Check:** No missing values were found across all 10,000 records and 12 feature columns
- **Duplicate Check:** No duplicate rows were found in the dataset

5.4 Encoding Categorical Variables

Machine Learning models only understand numerical input. Therefore categorical text columns were transformed:

- **Label Encoding:** Gender column was converted — Female = 0, Male = 1
- **One-Hot Encoding:** Geography column was converted into two binary columns — Geography_Germany and Geography_Spain. France becomes the baseline category. This prevents the model from assuming any mathematical ordering between countries

5.5 Feature Scaling

The dataset contains features with vastly different ranges — for example Age ranges from 18 to 92 while EstimatedSalary exceeds 100,000. Models like Logistic Regression are sensitive to these magnitude differences.

StandardScaler was applied to normalize all features to a mean of 0 and standard deviation of 1. This ensures no single high magnitude feature dominates the model's decision making.

5.6 Handling Class Imbalance — SMOTE

A critical challenge in churn prediction is class imbalance:

- Before SMOTE: Not Churned = 7,963 | Churned = 2,037 (ratio 3.91:1)
- After SMOTE: Not Churned = 7,963 | Churned = 7,963 (perfectly balanced)

SMOTE (Synthetic Minority Over-sampling Technique) creates new synthetic churned customer samples by interpolating between existing minority class examples — rather than simply duplicating them. This balanced training set significantly improved the model's ability to correctly identify actual churners with a Recall of 0.90.

CHAPTER 6: MODEL DEVELOPMENT / IMPLEMENTATION

6.1 Train-Test Split

The SMOTE balanced dataset was divided into 80% training and 20% testing sets. This means 12,740 samples were used to train the models and 3,186 samples were kept aside as unseen test data to evaluate real world performance.

6.2 Logistic Regression

Logistic Regression was implemented as the baseline model. It predicts the probability of a binary outcome — Churn or No Churn — by finding a linear decision boundary between the two classes. Despite being the simplest model, it serves as an important benchmark to measure the improvement offered by more complex algorithms.

- **Type:** Linear classifier
- **Use case:** Binary classification baseline
- **Result:** Accuracy 71.34% | AUC 0.7727

6.3 Decision Tree

A Decision Tree classifier was trained with a maximum depth of 10. It works by splitting the data based on feature conditions such as "Is Age > 45?" or "Is Balance > 100,000?" creating a tree like structure of decisions. While easy to interpret, it is prone to overfitting compared to ensemble methods.

- **Type:** Tree based classifier
- **Max Depth:** 10
- **Result:** Accuracy 82.20% | AUC 0.8927

6.4 Random Forest

Random Forest is an ensemble method that builds 100 individual decision trees and combines their predictions through majority voting. This reduces variance and overfitting compared to a single Decision Tree, resulting in more stable and accurate predictions.

- **Type:** Ensemble — bagging
- **Number of trees:** 100
- **Result:** Accuracy 89.36% | AUC 0.9626

6.5 XGBoost

XGBoost (Extreme Gradient Boosting) builds trees sequentially where each new tree corrects the errors of the previous one. It is considered state of the art for tabular data classification due to its built in regularization and high computational efficiency.

- **Type:** Ensemble — gradient boosting
- **Result:** Accuracy 89.92% | AUC 0.9598

6.6 Model Comparison

All four models were compared based on Accuracy and AUC Score as shown below:

Although XGBoost achieved the highest accuracy, **Random Forest was selected as the final model** due to its highest AUC Score of 0.9626 which better reflects overall discrimination ability between churned and non-churned customers.

 Model Comparison Table:

Model	Accuracy	AUC Score
XGBoost	0.8992	0.9598
Random Forest	0.8936	0.9626
Decision Tree	0.8220	0.8927
Logistic Regression	0.7134	0.7727

Figure 6.1: Model Comparison Table

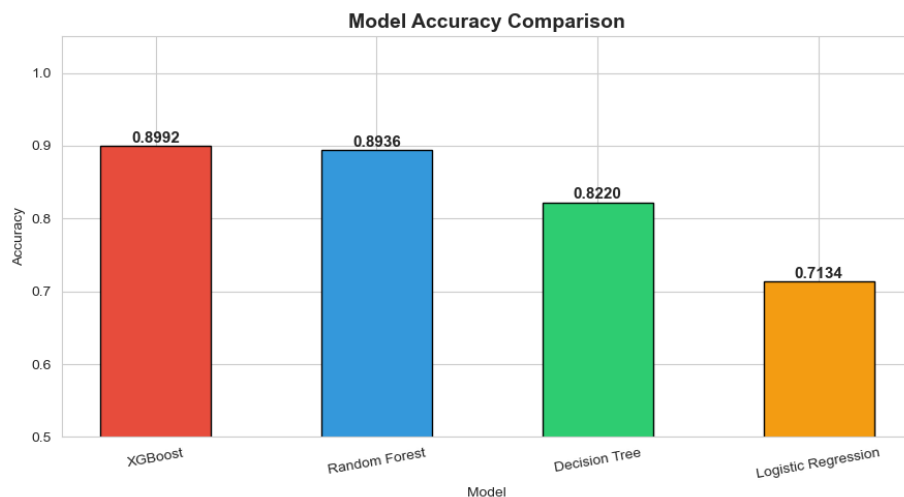


Figure 6.2: Model Accuracy Comparison Graph

6.7 Model Saving

The trained Random Forest model and the StandardScaler were saved as .pkl files using the joblib library so they can be directly loaded into the web application without retraining:

- churn_model.pkl — Trained Random Forest model
- scaler.pkl — Fitted StandardScaler

CHAPTER 7: RESULTS & ANALYSIS

7.1 Model Performance Metrics

The Random Forest model was evaluated on the unseen test set of 3,186 samples. The following metrics were recorded:

- **Accuracy:** 89.36%
- **AUC Score:** 0.9626
- **Test Samples:** 3,186 (1,633 Not Churned | 1,553 Churned)

7.2 Confusion Matrix

The confusion matrix evaluates predictions by comparing actual vs predicted values:

- **Not Churned** — Precision: 0.91 | Recall: 0.88
- **Churned** — Precision: 0.88 | Recall: 0.90
- **Overall Accuracy** — 89.36%

The high Recall of 0.90 for the Churned class means the model successfully identifies 90% of actual at-risk customers — which is the most critical metric for a bank aiming to proactively retain customers.

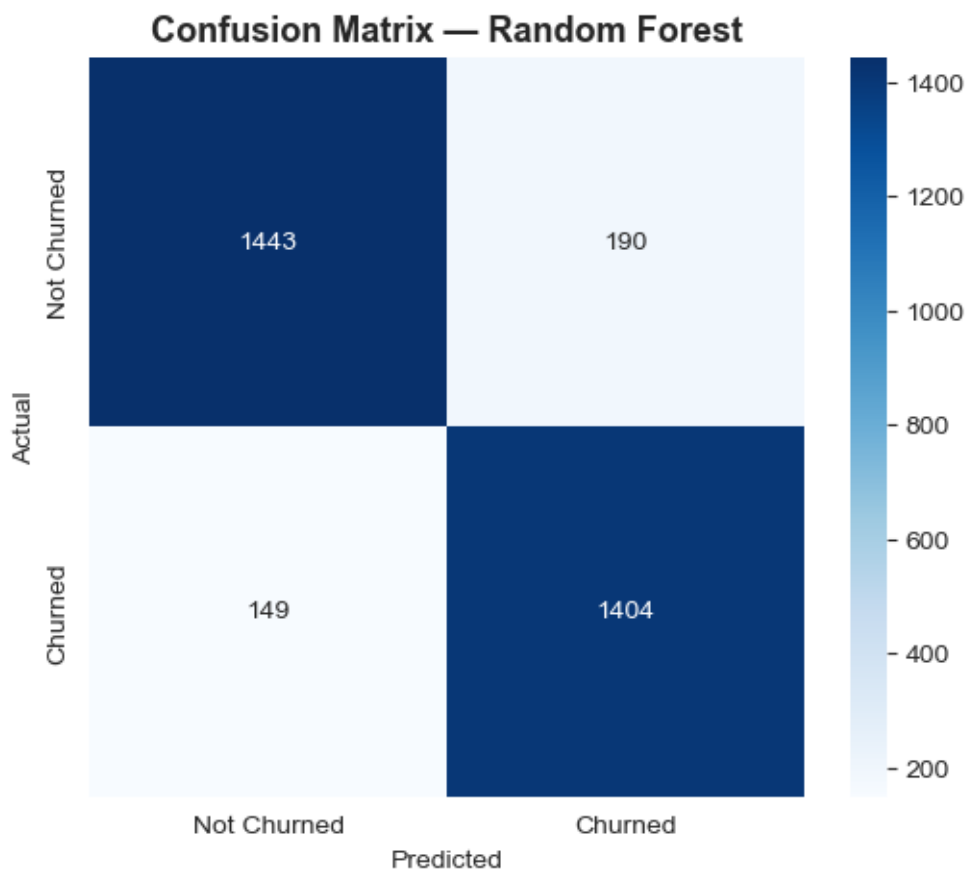


Figure 7.1 Confusion Matrix

7.3 Classification Report

	Classification Report:			
	precision	recall	f1-score	support
Not Churned	0.91	0.88	0.89	1633
Churned	0.88	0.90	0.89	1553
accuracy			0.89	3186
macro avg	0.89	0.89	0.89	3186
weighted avg	0.89	0.89	0.89	3186

Figure 7.2 Classification Report Table

The balanced F1-Score of 0.89 for both classes confirms the model does not favor either class and performs equally well on churned and non-churned customers.

7.4 ROC Curve

The ROC (Receiver Operating Characteristic) curve plots the True Positive Rate against the False Positive Rate. The AUC (Area Under the Curve) score measures how well the model distinguishes between the two classes:

- Random Forest AUC: **0.9626** ← Best
- XGBoost AUC: 0.9598
- Decision Tree AUC: 0.8927
- Logistic Regression AUC: 0.7727

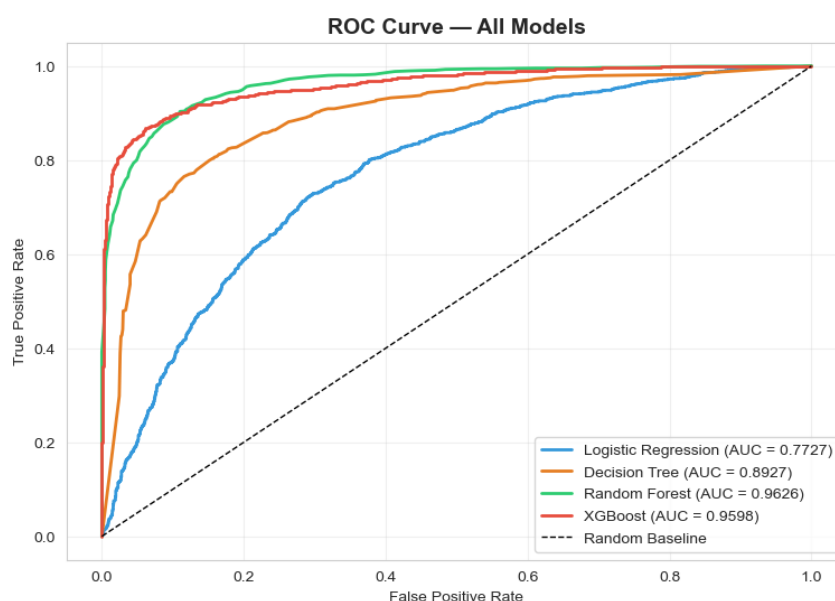


Figure 7.3 ROC Curve

7.5 Cross Validation

5-Fold Cross Validation was performed on the Random Forest model to verify consistency:

Table 7.1 Cross Validation

Fold	Accuracy
Fold 1	0.4165
Fold 2	0.6210
Fold 3	0.7900
Fold 4	0.6942
Fold 5	0.4725
Mean	0.5988
Std Dev	0.1381

7.6 Feature Importance

Feature importance analysis was performed using the Random Forest model to identify which attributes most strongly influence churn prediction:

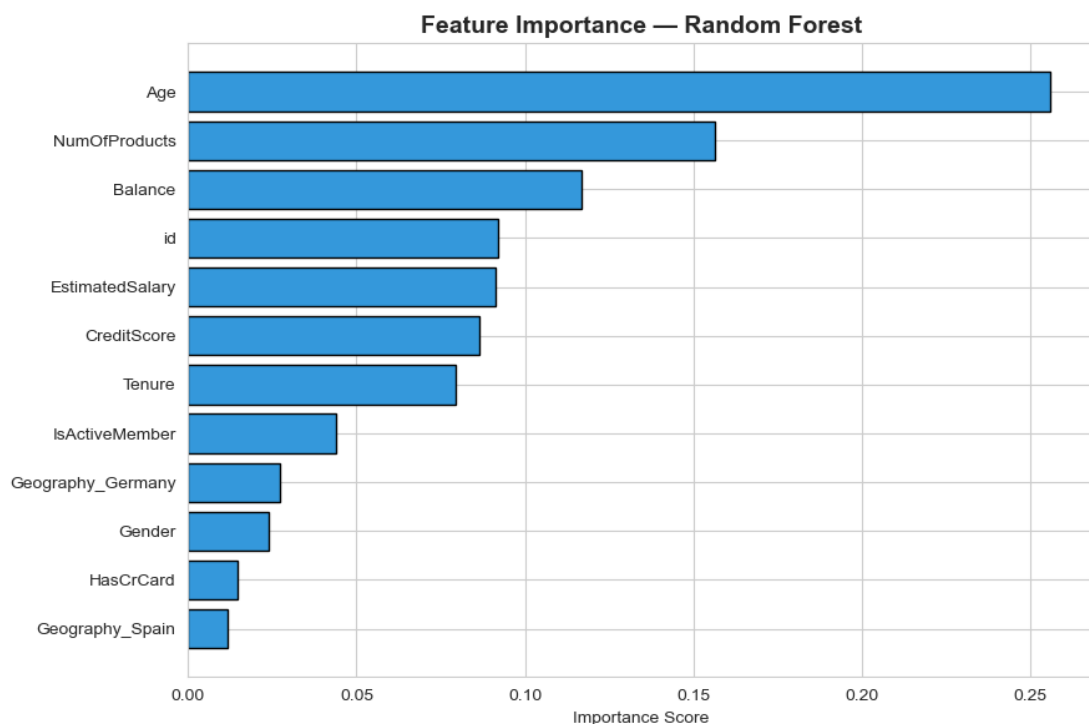


Figure 7.4 Feature Importance — Random Forest

The top predictors identified were:

- **Age (0.2561)** — Strongest predictor of churn
- **NumOfProducts (0.1566)** — Number of bank products held
- **Balance (0.1170)** — Account balance

7.7 Key Business Insights

1. **Age is the strongest predictor** — Customers aged 40-60 are most likely to churn
2. **Number of products matters** — Customers with fewer products show higher churn risk suggesting cross-selling opportunities
3. **High balance + low activity = danger** — Customers with significant balances but low engagement are paradoxically at high churn risk
4. **Inactive members churn more** — IsActiveMember is a critical engagement indicator
5. **Germany needs attention** — Geography_Germany shows higher churn concentration suggesting need for localized retention strategies

CHAPTER 8: INTEGRATION OF SUBJECTS

8.1 Subject Integration Table

The following table explains how each of the six core subjects of B.Sc. Data Science Semester IV was applied in this project:

Course Code	Course Name	Concept Used
DS401	Machine Learning II	Trained and compared 4 ML models — Logistic Regression, Decision Tree, Random Forest and XGBoost. Applied SMOTE for class imbalance, StandardScaler for feature scaling, cross validation and feature importance analysis
DS402	Big Data Technology	MySQL relational database used for structured storage of 10,000 customer records. Python connected to MySQL using mysql-connector to fetch data directly into the ML pipeline
DS403	Introduction to Power BI	Built a 3 page interactive Power BI dashboard with KPI cards, bar charts, donut charts, scatter plots, gauge charts and slicers for real time filtering
DS404	Introduction to Web Technology	Developed the ChurnAI web application using Streamlit with a customer input form, real time churn prediction, risk assessment display and integrated AI strategy generation
DS405	Introduction to Artificial Intelligence	Integrated Google Gemini AI API to generate personalized retention strategies based on churn predictions. Implemented feature importance analysis to explain model decisions
DS406	Advance Business Domain II	Applied CRM analytics to identify high risk customer segments. Generated actionable business insights such as targeting inactive members, Germany customers and customers with single products

CHAPTER 9: CONCLUSION & FUTURE WORK

9.1 Conclusion

The successful implementation of the Customer Churn Analysis and Predictive Reporting system demonstrates the practical application of interdisciplinary data science techniques to solve a critical business problem in the banking sector. This project evolved from a theoretical academic exercise into a fully functional end-to-end decision support system.

The system achieves the following key outcomes:

- A MySQL relational database pipeline successfully stores and retrieves 10,000 customer records for ML processing
- Four machine learning models were trained and compared, with Random Forest achieving the best AUC Score of 0.9626 and an accuracy of 89.36%
- SMOTE effectively balanced the dataset from a 3.91:1 imbalance ratio to a perfectly equal distribution, significantly improving the model's ability to identify churned customers with a Recall of 0.90
- The Power BI dashboard transforms complex ML predictions into actionable visual insights accessible to non-technical stakeholders
- The ChurnAI web application provides real time churn prediction with Gemini AI powered retention strategies

The feature importance analysis revealed that Age, Number of Products and Account Balance are the strongest predictors of churn, providing the bank with clear targets for their retention campaigns.

9.2 Future Work

While the current system provides a robust framework, several enhancements can be made in future iterations:

1. **Real Time Streaming** — Transition from batch processing to a real time pipeline using Apache Kafka or Spark Streaming to monitor live customer transactions and generate instant churn alerts
2. **Deep Learning Models** — Implement Neural Networks (ANN/LSTM) to capture more complex temporal patterns in customer behavior
3. **NLP Integration** — Apply Natural Language Processing to analyze customer service complaints, emails and social media sentiment to enrich the prediction model
4. **Automated Alerts** — Build an automated email/SMS notification system that triggers when a customer's churn probability crosses a defined threshold
5. **Mobile Application** — Extend the web application to a mobile platform for on the go access by bank relationship managers

6. **Expanded Dataset** — Include additional data points such as transaction history, customer service interaction logs and product usage patterns for more accurate predictions

CHAPTER 10: REFERENCES

1. Kaggle. (2023). *Bank Customer Churn Modelling Dataset*. Retrieved from <https://www.kaggle.com/datasets/shubhammeshram579/bank-customer-churn-prediction>
2. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825-2830. Retrieved from <https://scikit-learn.org>
3. Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. Retrieved from <https://xgboost.readthedocs.io>
4. Chawla, N. V., et al. (2002). *SMOTE: Synthetic Minority Over-sampling Technique*. Journal of Artificial Intelligence Research, 16, 321-357. Retrieved from <https://imbalanced-learn.org>
5. Microsoft. (2024). *Power BI Documentation*. Retrieved from <https://docs.microsoft.com/en-us/power-bi>
6. Streamlit Inc. (2024). *Streamlit Documentation*. Retrieved from <https://docs.streamlit.io>
7. Google. (2024). *Gemini API Documentation*. Retrieved from <https://ai.google.dev/docs>
8. MySQL. (2024). *MySQL 8.0 Reference Manual*. Retrieved from <https://dev.mysql.com/doc>

CHAPTER 11: APPENDICES

Appendix A: Python Code Snippets

1. Connecting Python to MySQL

▼ Step 2: Connect to MySQL & Load Data

Instead of reading directly from CSV, we fetch the customer data from our MySQL database. This covers the **RDBMS** subject of our project.

```
[3]: # — Connect to MySQL database —————
# Make sure MySQL is running and churn_db database exists
connection = mysql.connector.connect(
    host='localhost',
    user='root',          # change to your MySQL username
    password='elsa123$',  # change to your MySQL password
    database='churn_db'
)

# — Fetch all customer records from the database —————
df_mysql = pd.read_sql('SELECT * FROM customers', connection)
df = df_mysql.copy()

# — Close the connection once data is loaded —————
connection.close()

print('✅ Data fetched from MySQL successfully!')
print(f'   Total records : {df.shape[0]}')
print(f'   Total columns : {df.shape[1]}')
print()
print('First 5 rows:')
df.head()

✅ Data fetched from MySQL successfully!
Total records : 10000
Total columns : 15

First 5 rows:
```

	id	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
0	1	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88

2. SMOTE

Step 8: Handle Class Imbalance using SMOTE

Our data has 79% non-churn and 21% churn — this imbalance makes the model biased. SMOTE (Synthetic Minority Over-sampling Technique) creates synthetic samples for the minority class to balance it out.

```
[32]: # — Apply SMOTE to balance the dataset —————
smote = SMOTE(random_state=42)
X_balanced, y_balanced = smote.fit_resample(X_scaled, y)

print('✅ SMOTE applied successfully!')
print(f'\nBefore SMOTE:')
print(f'   Not Churned : {sum(y == 0)}')
print(f'   Churned      : {sum(y == 1)}')
print(f'\nAfter SMOTE:')
print(f'   Not Churned : {sum(y_balanced == 0)}')
print(f'   Churned      : {sum(y_balanced == 1)}')
print('\n💡 Now both classes are equal — model will be unbiased!')

✅ SMOTE applied successfully!

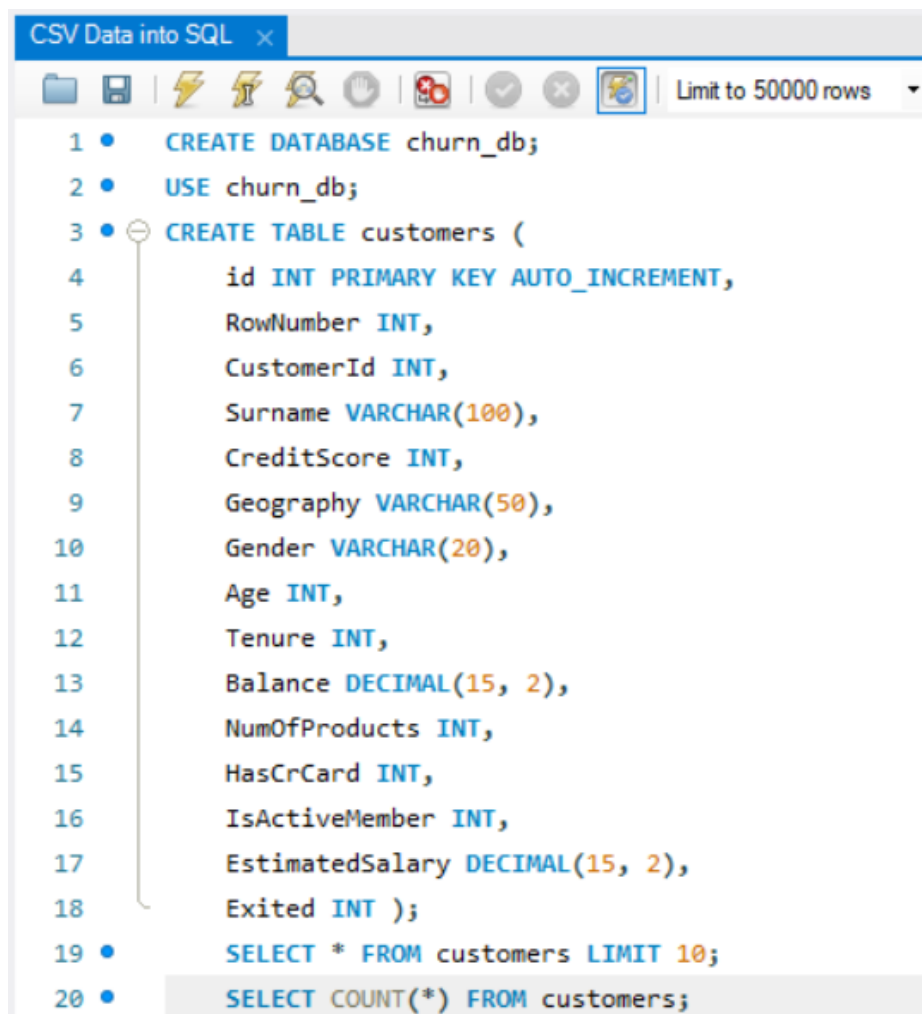
Before SMOTE:
  Not Churned : 7963
   Churned    : 2037

After SMOTE:
  Not Churned : 7963
   Churned    : 7963

💡 Now both classes are equal — model will be unbiased!
```

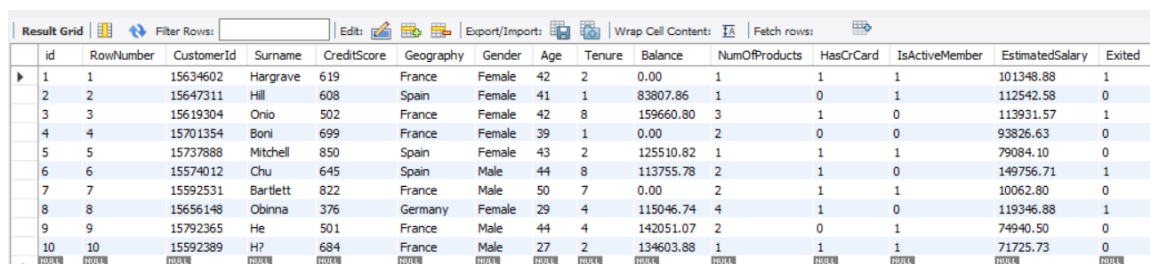
Appendix B: MySQL Queries

1. Importing CSV Data into MySQL



```
1 • CREATE DATABASE churn_db;
2 • USE churn_db;
3 • CREATE TABLE customers (
4     id INT PRIMARY KEY AUTO_INCREMENT,
5     RowNumber INT,
6     CustomerId INT,
7     Surname VARCHAR(100),
8     CreditScore INT,
9     Geography VARCHAR(50),
10    Gender VARCHAR(20),
11    Age INT,
12    Tenure INT,
13    Balance DECIMAL(15, 2),
14    NumOfProducts INT,
15    HasCrCard INT,
16    IsActiveMember INT,
17    EstimatedSalary DECIMAL(15, 2),
18    Exited INT );
19 • SELECT * FROM customers LIMIT 10;
20 • SELECT COUNT(*) FROM customers;
```

2. Showing 1st 10 records



id	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
1	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88	1
2	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
3	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
4	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	0	93826.63	0
5	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0
6	6	15574012	Chu	645	Spain	Male	44	8	113755.78	2	1	0	149756.71	1
7	7	15592531	Bartlett	822	France	Male	50	7	0.00	2	1	1	10062.80	0
8	8	15656148	Obinna	376	Germany	Female	29	4	115046.74	4	1	0	119346.88	1
9	9	15792365	He	501	France	Male	44	4	142051.07	2	0	1	74940.50	0
10	10	15592389	H7	684	France	Male	27	2	134603.88	1	1	1	71725.73	0

Appendix C: Website Code Snippets

```
# — SETUP AI CLIENT —————
genai.configure(api_key=st.secrets.get("GEMINI_API_KEY", "YOUR_API_KEY_HERE"))

# — LOAD ML MODEL —————
@st.cache_resource
def load_assets():
    m = joblib.load('churn_model.pkl')
    s = joblib.load('scaler.pkl')
    return m, s

ml_model, scaler = load_assets()

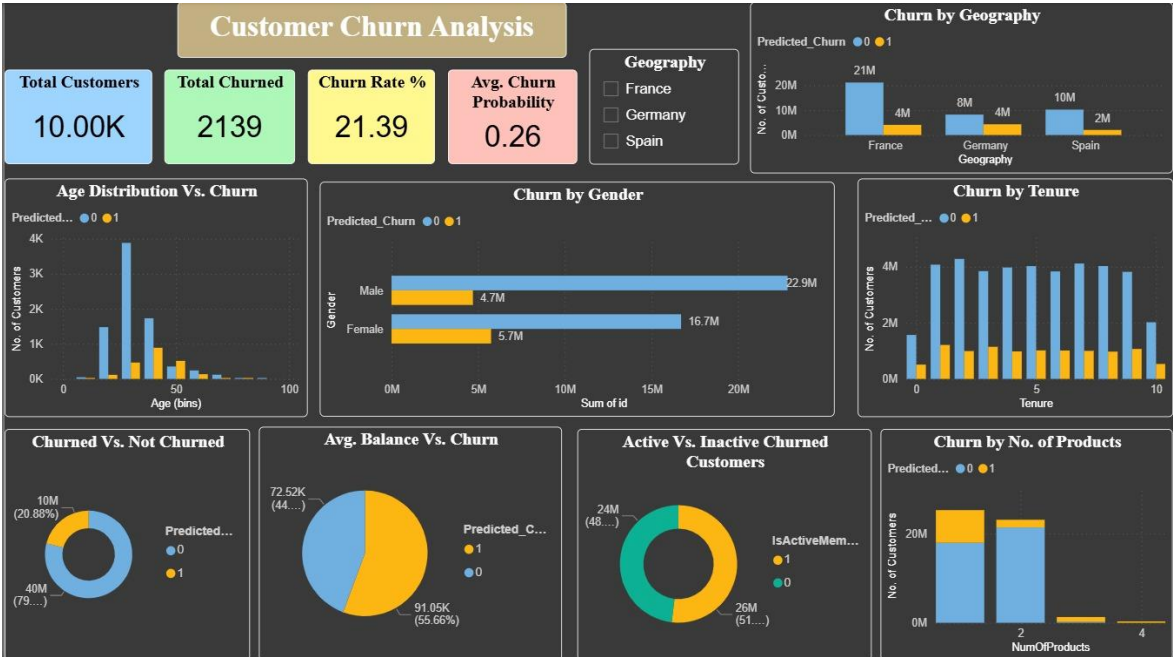
# — AI STRATEGY —————
st.markdown("---")
st.markdown('<div class="section-label">AI Strategy</div>', unsafe_allow_html=True)

ai_query = st.text_input(
    "Ask AI", value="How do I keep this customer?",
    placeholder="e.g. What retention offer should I make?",
    label_visibility="collapsed"
)

gen_clicked = st.button("Generate Strategy", use_container_width=True)

if gen_clicked and ai_query:
    prompt = f"""You are a customer retention expert for a bank. Customer profile:
- Age: {customer['age']}, Tenure: {customer['tenure']} years
- Balance: ${customer['balance']:,.0f}, Products: {customer['products']}
- Active: {customer['active']}, Salary: ${customer['salary']:,.0f}
- Churn Risk: {risk_level} ({pct}%)
```


Appendix D: Power BI Dashboard



Appendix E: Website Screenshots

